A Lean Audit of the IUT III Corollary 3.12 Corridor

IUT Lean formalization project

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Abstract

We report on a first Lean formalization of the logical corridor from IUT III, Theorem 3.11 to Corollary 3.12. The formalization is deliberately neutral with respect to the correctness of IUT as a proof of the abc conjecture. It isolates the ordered-real comparison required at the end of Corollary 3.12, tracks the finite q^{j^2} and label-quotient phenomena that enter the public dispute, and records exactly which source-facing construction hypotheses are still external.

The current code proves conditional theorems. If the hull/log-volume upper-ray comparison supplies $-|\log(q)| \leq -|\log(\Theta)|$, and if the displayed hypothesis $-|\log(\Theta)| \leq C_\Theta|\log(q)|$ is supplied with $|\log(q)| > 0$, then Lean proves $C_\Theta \geq -1$. If the upper-ray membership is strict, Lean proves $C_\Theta > -1$. In the boundary case $C_\Theta = -1$, Lean proves equality of the q-pilot reading and the Θ-hull boundary, and then classifies the local Frobenioid p_v^N-shift behavior under the actual upper-ray quotient. This does not prove Corollary 3.12 from Mochizuki's published hypotheses; it identifies the remaining mathematical construction problem.

After a source audit against IUT I–IV, Mochizuki's 2026 LeanForm report, and Scholze–Stix, the status is unchanged in kind but sharper in formulation: the final ordered-real implication is formalized, and several collapse-prone comparison levels are separated, but the code still assumes the source construction of the multiradial/HDD/SHE/IPL data that IUT III assigns to Theorem 3.11 and its reorganization.

1 Aim and Sources

The project formalizes a narrow and contested part of Mochizuki's inter-universal Teichmuller theory. The source base is the local copy of IUT I–IV, Mochizuki's 2026 report on formalization, and the Scholze–Stix note. The current target is the portion that Mochizuki's report calls Stage 1:

$$\text{IUT III, Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{IUT III, Corollary 3.12.}$$

The notation “3.11.5” is not a statement in IUT III. It is Mochizuki's 2026 name for the reorganized boundary where the hull-plus-determinant operation is moved before the final simultaneous comparison of the Θ-pilot and the q-pilot.

The formalization is source-facing but not source-complete. It does not construct Hodge theaters, Frobenioids, log-shells, prime strips, or theta links from first principles. Instead, these are represented by typed records with named hypotheses. The point of the current work is to test whether the claimed formal corridor is coherent once every comparison level, label quotient, ordered real line, hull boundary, and quotient collapse is made explicit.

2 Framework and Tooling

The repository is organized as a Lean verification project, not as a prose annotation project. The source papers in `docs/` are treated as the mathematical source of truth. The formalization then moves through three layers:

$$\begin{gathered}\text{source papers}\\ \Downarrow \\ \text{Lean definitions, records, and theorems}\\ \Downarrow \\ \text{this paper draft.}\end{gathered}$$

The paper draft records only the current mathematical state of the formalization. It is not a research log.

The Lean code uses typed interfaces to prevent comparison-level drift. A datum that is merely a representative j^2-coordinate computation has a different type from a balanced full-label datum, and both differ from the final hull/log-volume comparison datum. Likewise, ordered real-line transports, Hodge/SHE preservation records, upper-ray membership, and quotient-collapse audits are separate structures. This is the main anti-collapse mechanism in the framework: a proof cannot silently use a raw coordinate equality where the Corollary 3.12 corridor asks for a hull/log-volume comparison.

The module roles are:

- `IUTStage1SourceCore`: core real-line, pilot, hull, quotient, and C_Θ algebra.
- `IUTStage1Remark312Absorption`: Remark 3.12.2(v) absorption interfaces.
- `IUTStage1Source`: finite labels, Gaussian data, SHE/Hodge interfaces, and route theorems.
- `IUTStage1Experiments`: compiled endpoint checks and diagnostic theorems.

The experiment file is not a substitute for proof. It is a curated surface that forces the important route theorems to elaborate together and exposes Boolean/report endpoints such as whether the present stage settles the dispute. At present that endpoint is deliberately false.

The build discipline is simple. Each mathematical increment is checked by Lean via `lake build Iut.Stage1.IUTStage1Experiments`; the paper is rebuilt from this LaTeX source; and the code is scanned for explicit proof placeholders such as `sorry`, `admit`, and `axiom`. The long-term collaboration plan is compatible with Apodixis: the current typed route endpoints and missing-obligation records give a natural surface for a collaboration tool to show which mathematical interfaces are proved and which remain open.

3 Dispute Interface

In IUT III, Step (x) treats the procession-normalized log-volume as unchanged under (Ind1) and (Ind2), while (Ind3) supplies a one-sided upper-semi comparison. Step (xi) then passes through multiradiality, IPL, SHE, holomorphic hulls, determinants, and log-volume to place the q-pilot reading in the signed-log upper ray determined by the Θ-pilot:

$$-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|}.$$

The active source-facing Corollary 3.12 route does not use equality of possible-image regions across an (Ind3) generator. Such an equality exists only in separate quotient bookkeeping interfaces as an explicit stronger hypothesis; the route uses the one-sided upper-semi corridor as its Step (x) input. The current Theorem 3.11 source record now bundles structured IPL/SHE/APT input, coric multiradial output, possible theta-pilot images, the 0-column/1-column pilot distinction, distinct log-shell treatments, and the Hodge-theater history guard. This is still a source interface, not a construction of Theorem 3.11. The IPL layer has been separated into a typed log-volume transport datum: it carries the source and target Hodge-theater sides, identifies the prime-strip link endpoints with the input and output prime strips, records equality of the transported real log-volume, and retains the no-history-identification guard. It is not yet a construction of the full log-link. The current SHE layer now has a synchronization constructor: equality of the canonical one-label Hodge–Arakelov theta-value degree constructs the factored square/full-label preservation package used by the route. This is a Stage 1 source constructor for the available finite-label shadow of SHE, not yet the full simultaneous-holomorphic-expressibility theorem. The hull/determinant layer now has a source constructor attached to the Theorem 3.11 record. It reads the algorithmic certificate and SHE alignment from that record, keeps the record's theta possible-image union tied to the hull endpoint, and derives the determinant-bound comparison $q_{\text{signed}} \leq \theta_{\text{signed}}$. The bridge used by this source data can now be constructed from a displayed common hull containing the target union and a determinant/log-volume bound for that hull; the analytic construction of the hull and determinant remains external source data. The strongest Hodge–Arakelov packet-target route no longer takes the target-side log-volume equality or q-pilot positivity as independent inputs: target equality is derived from the SHE synchronization constructor, and positivity is read from the hull/determinant constructor. The source-side chart calibration is no longer a raw equality input: it is a full-label theta/Hodge calibration certificate, from which equality of the cusp log-volume compatibility records is derived extensionally. The final numerical step is elementary:

$$-|\log(q)| \leq -|\log(\Theta)| \leq C_\Theta |\log(q)|, \qquad |\log(q)| > 0 \quad \Longrightarrow \quad C_\Theta \geq -1.$$

Scholze–Stix object to the comparison because the Θ-side data carries j^2-scaled degrees, while a naive identification of ordered real lines appears to remove the room needed for these scalars. The formalization therefore separates four comparison levels:

representative j^2 data
balanced sign-compatible data
averaged cusp-label data
hull/log-volume data.

The final route consumes only the hull/log-volume comparison level. Lean rejects the balanced sign-compatible level as sufficient for the final comparison.

4 Source Audit

The local sources impose the following constraints on the formalization. IUT I constructs the initial Θ-data and the $\Theta^{\pm\text{ell}}$NF-Hodge theaters. It is not merely a notation layer: the paper's central point is that the Hodge theaters separate the additive and multiplicative combinatorial dimensions of a number field. IUT II supplies the Hodge–Arakelov theta evaluation, theta values at ℓ-torsion points, Gaussian Frobenioids, conjugate synchronization, and the multiradial/coric phenomena needed later. The current code models only finite label and Gaussian degree shadows of this material.

IUT III is the direct source for the formal corridor. Step (x) says that procession-normalized log-volumes are invariant under (Ind1) and (Ind2), while (Ind3) becomes an upper-semi inequality after applying log-volume. Step (xi) uses the multiradial construction, IPL, SHE, holomorphic hulls, determinants, and the 1-column log-Kummer correspondence to put the q-pilot log-volume in the signed-log upper ray $\mathbb{R}_{\leq -|\log(\Theta)|}$. Step (xii) warns that replacing global realified Frobenioids by local Frobenioids loses calibration because local objects admit p_v^N-type endomorphisms. Remark 3.12.2 further stresses that the argument depends on weakened data, distinct q/Θ labels, upper-semi indeterminacy, and the asymmetry between the 0-column Θ-pilot and the 1-column q-pilot. The current Step (x) packet source reflects this asymmetry by packaging the 1-column q-pilot non-interference, Kummer/forgetting product preservation, source packet calibration, and packet-normalized target calibration in a single realified-Frobenioid log-Kummer source object.

IUT IV uses the resulting log-volume estimate as input to elementary log-volume and Diophantine inequalities. Our IUT IV code is therefore only conditional algebra at present: it records how later estimates consume a Corollary 3.12-shaped bound, but it does not prove that bound from the full arithmetic constructions.

Mochizuki's 2026 report explicitly recommends Stage 1 as the implication *IUT* III, Theorem 3.11 $\Rightarrow$ Corollary 3.12, with a reorganized intermediate "3.11.5" obtained by moving the hull-plus-determinant operation before the final simultaneous comparison. The same report states that the current official code is skeletal. Our formalization is best read as a more diagnostic Stage 1 interface: it does not replace Stage 2, because it has not constructed Theorem 3.11.

Scholze–Stix argue that, after simplification, the comparison loses the j^2-scaled theta contribution and becomes essentially contentless. The Lean code therefore treats the j^2-representative level, sign quotient, averages, square/full-label preservation, ordered real-line transports, and hull/log-volume endpoint as different types. This confirms that the final Corollary 3.12 endpoint cannot be obtained from the balanced sign-compatible level alone. It does not prove that Mochizuki's source constructions supply the missing higher-level data.

5 Formalized Corridor

The relevant implementation lives in four Lean modules:

- `Iut/Stage1/IUTStage1SourceCore.lean`
- `Iut/Stage1/IUTStage1Remark312Absorption.lean`
- `Iut/Stage1/IUTStage1Source.lean`
- `Iut/Stage1/IUTStage1Experiments.lean`

5.1 Finite label model

The code defines a finite $\mathbb{F}_\ell$-label model, the quotient $|F_\ell|$ consisting of the zero class together with $\mathbb{F}_\ell^\times/\{\pm 1\}$, and the canonical half-range representatives

$$j = 0, \ldots, (\ell - 1)/2.$$

The Stage 1 label model is now connected to the bad-local theta-root data of IUT I, Definition 3.1(e),(f). A new source package consumes the quotient-$\mathbb{Z}$ data of a bad local theta-root model and derives the Stage 1 full label from its canonical coordinate. It also carries the canonical-generator-up-to-sign witness. In the concrete $Z/\ell Z$ quotient case this derived full label is exactly the nonzero canonical full label corresponding to coordinate 1. It proves, for the descended full-label degree model:

$$\deg(\Theta_j) = j^2 \deg(q)$$

for the degree-level Gaussian endpoint, with sign invariance on the full-label quotient. It also proves that $j = 1$ and $j = 2$ give distinct degrees exactly when $\deg(q) \neq 0$. Thus the q^{j^2} phenomenon is present in the model; it is not hidden inside a scalar identification.

The raw coordinate square-weight profile on $\mathbb{Z}/\ell\mathbb{Z}$ is kept as a diagnostic object and is not identified with the descended $|F_\ell|$ theta exponent. The coordinate average, the nonzero-coordinate average, and the absolute-label average are all formalized separately. Lean proves the exact denominator relations and shows that these averages differ strictly for nonzero environment degree. This prevents silently replacing the paper's ℓ-denominator convention by an auxiliary nonzero-only average.

5.2 Indeterminacy and upper-semi route

The Step (x) corridor is represented as:

$$A_0 \quad = \quad A_1 \quad = \quad A_2 \quad \leq U_{\mathrm{Ind3}},$$

where A_0, A_1, A_2 are the before-, after-(Ind1)-, and after-(Ind2)-averaged log-volumes, and U_{Ind3} is the (Ind3) upper-semi target. From this data and the hull/determinant identifications, the code constructs the Step (xi) upper-ray datum:

$$q_{\mathrm{avg}} \leq \Theta_{\mathrm{hull}}.$$

The same route constructs the two q-pilot readings in Step (xi-g) and proves that they are equal to the same fixed real value.

5.3 SHE and Hodge-descent boundary

The current SHE layer is conditional. It has typed records for:

structured SHE square-weight transport
factored square/full-label preservation
Hodge-descent packet transport
nonarchimedean (Ind3) source-target alignment.

Given these inputs, Lean constructs the weighted theta comparison consumed by the final route:

$$q_{\mathrm{signed}} \leq \Theta_{\mathrm{signed}}.$$

The nonarchimedean (Ind3) entry, factored square/full-label preservation, and Hodge-descent packet transport now compose directly to this comparison. With the q-pilot positivity condition and the displayed C_Θ-bound, the same composed route instantiates the signed C_Θ endpoint and proves $C_\Theta \geq -1$. The instantiated endpoint also exposes the strict and boundary alternatives: strict source comparison gives $C_\Theta > -1$, while $C_\Theta = -1$ forces equality of the signed q- and Θ-readings and makes the Θ-reading negative. These facts are now also packaged as a single route-level alternative: boundary equality with negative Θ-reading, or strict $C_\Theta > -1$. The code also proves a Gaussian-to-factored-SHE bridge at the present finite degree level. In particular, equality of the source and target environment degrees gives full-label value preservation; nonzero-label target bounds are enough because the zero label has square weight zero. The newest route uses this bridge before applying the nonarchimedean endpoint: Gaussian degree evaluations with preserved environment degree now construct the factored square/full-label preservation package consumed by the nonarchimedean C_Θ dichotomy. This removes one primitive preservation input from that route, but it still assumes the nonarchimedean (Ind3) alignment and the Hodge-descent/SHE source audit. There is also an identity-coordinate specialization. In that case equality of Gaussian degree at the canonical full label 1 implies preservation of the environment degree, so the nonarchimedean C_Θ dichotomy no longer takes environment-degree preservation as an independent hypothesis. The Hodge–Arakelov/Gaussian bridge is now source-facing at the current degree level. A theta-value evaluation source carries the bad-local theta-root label data and a theta-monoid degree d, and constructs the Gaussian degree evaluation

$$\deg_{\mathrm{gau}}(\lambda) = e(\lambda)\, d,$$

where $e(\lambda)$ is the absolute-label theta exponent. The zero label has degree 0, and the canonical label 1 has degree d. Thus equality at the canonical label derives equality of the environment degree and feeds the factored SHE square/full-label obligations. The raw nonarchimedean entry alignment has also been refactored through a source-facing nonarchimedean upper-semi object equipped with 1-column q-pilot log-Kummer data. The upper-semi inequality remains the Step (x) input; the 1-column datum records the q-pilot column and log-Kummer non-interference before converting to the raw real equalities consumed by the current route. For the source side of this compatibility, the code can now use the existing finite tensor-packet chain: Kummer transfer holomorphicF $\to$ holomorphicD and mono-analytic forgetting holomorphicD $\to$ $D^{\vdash}$ preserve product log-volume. Thus the source-side upper-semi equality may be derived from this chain rather than postulated as a bare equality. The newest source package attaches this finite Kummer/forgetting chain to the bad-local theta-root cusp-label source. It carries the canonical nonzero $|F_\ell|$-label and derives the nonarchimedean (Ind3) source equality from product-log-volume preservation along

$$\text{holomorphicF} \to \text{holomorphicD} \to D^{\vdash}.$$

The packet local object, local entry source, mono-analytic product, and holomorphic source product are now packaged as one source-side log-Kummer packet calibration object. The object itself is still supplied source data, but the equality from entry source to (Ind3) source is derived from it together with the Kummer and forgetting maps. The source calibration can now also be constructed from the explicit packet-local/source alignment, the packet-local (Ind3)-source alignment, and the two product-preservation maps. Thus the route no longer has to accept the holomorphic source-product equality as a primitive field when these Step (x) alignments are available. The source and target packet calibrations are now joined by a finite log-Kummer packet-correspondence source carrying the q-pilot column and non-interference guard. Its target face carries the common packet-normalized capsule value, and Lean derives the direct theta-to-entry, theta-to-packet, and entry-to-(Ind3)-target equalities consumed by the upper-semi route. The target calibration no longer contains an independent real representative for this common value: all target equalities are stated directly against the packet-normalized capsule log-volume. This removes one artificial identification layer from the Step (x) interface. The target face has also been refined by a vertical-IQ source object, following IUT III, Proposition 3.10(ii). The theta average, the (Ind3)-target value, and the local entry target are represented as points of one log-Kummer IQ-target whose values agree with the packet-normalized point. The packet-target calibration is then constructed from this vertical-IQ source, so the corresponding theta-to-entry and entry-to-(Ind3)-target equalities are derived from common-target compatibility rather than passed as separate target-calibration inputs. This object is guarded by the exact $FMOD$ side of Proposition 3.10(iii) and Remark 3.10.1: the one-sided $Fmod$ upper-semi mode is classified separately and is not accepted as an exact vertical-IQ target calibration. The local entry target can now also be constructed from the packet-normalized capsule value itself. A packet-normalized entry-target source builds a nonarchimedean upper-semi entry whose target finite log-volume is definitionally the packet-normalized capsule value. The corresponding target calibration therefore no longer needs an independent proof for the entry-target equality; only the theta-average and (Ind3)-target identifications with the same packet-normalized value remain as inputs at this level. The latter can now be reduced further: if the Step (x) target-side (Ind3) alignment gives equality of the theta average with the (Ind3) target, then Lean derives the (Ind3)-target-to-packet-normalized equality from this alignment and the theta-average-to-packet-normalized equality. The packet-correspondence source itself can now be built from this reduced target data together with the source packet/product calibration. Thus the correspondence constructor no longer needs a completed target-calibration record when the target entry and theta-target alignment are available. It can also be built directly from the packet-local source alignments and the same target theta alignment. In this form both the source packet calibration and the target packet calibration are generated before the q-pilot non-interference endpoint is projected. The same reduction has been lifted through the realified-Frobenioid packet source: realified Kummer compatibility still supplies the holomorphicF $\to$ holomorphicD $\to D^{\vdash}$ transfer, while the target face is now generated from the packet-normalized entry target and theta-target alignment. The realified entry source and the theta-root attached realified entry source can also consume the vertical-IQ target source directly; their upper-semi endpoint then derives the theta-to-entry and entry-to-(Ind3)-target equalities through the common log-Kummer target rather than through separate target alignment inputs. This vertical-IQ source now reaches the integrated Hodge/SHE/IPL/hull route: the first-pass C_Θ dichotomy can be stated without the ordered-real-line target alignment and without the auxiliary packet-normalized $ZMod(\ell)$ route whenever the exact common IQ-target is supplied. The same route has a source-reduced form: packet-local/source alignment, entry-source/mono-analytic-product alignment, packet-local/(Ind3)-source alignment, and the realified Kummer/forgetting compatibilities construct the source calibration internally before the vertical-IQ target enters the final inequality route. This reduction now also exists at the Step (x) source-object level: an upper-semi packet-normalized entry, packet-local source alignments, realified Kummer/forgetting compatibility, and an exact vertical-IQ target construct one realified log-Kummer entry source before the Hodge/SHE/IPL/hull route is applied. The realified source can now also generate the source packet calibration from packet-local/source alignments before forming the correspondence; consequently the strongest realified Step (x) constructor no longer needs a completed source calibration as input. The first-pass Gaussian route also has a one-correspondence form: it consumes a finite log-Kummer packet-correspondence source and projects the source packet/product calibration and target packet-normalized calibration from that single object before entering the upper-semi corridor. The same correspondence now has a transfer-chain endpoint exposing the holomorphicF $\to$ holomorphicD $\to D^{\vdash}$ product log-volume equalities, the source equality with the (Ind3) real source, the target packet-normalized equalities, and q-pilot non-interference from the same object. The current strongest Step (x) entry source places this correspondence below the realified Frobenioid layer: the selected nonarchimedean upper-semi entry and the realified-Frobenioid packet source are bundled together, and Lean projects the Kummer transfer, mono-analytic forgetting transfer, source calibration, target packet calibration, and q-pilot non-interference guard from that single source object. There is now a reduced constructor for this entry source: entry membership, packet-local/source alignment, packet-local (Ind3)-source alignment, packet-normalized target data, and theta-target alignment construct the realified packet source internally and then derive the upper-semi inequality. The bad-local theta-root label source is now attached to this realified entry source. The corresponding Hodge–Arakelov route consumes one object carrying the canonical nonzero theta-root label, the selected upper-semi entry, and the realified-Frobenioid log-Kummer packet source before entering the C_Θ-dichotomy. This theta-root attached source now has the source-reduced vertical-IQ constructor: the theta-root label package is combined with the upper-semi packet-normalized entry, packet-local source alignments, realified Kummer/forgetting compatibility, and the exact vertical-IQ target. Lean then projects the canonical-label facts, transfer-chain product equalities, target packet calibrations, q-pilot non-interference, and the Step (x) upper-semi inequality from this single source object. The packet-local source side is now bundled as a named alignment object: packet-local/source equality, entry-source/mono-analytic-product equality, and packet-local/(Ind3)-source equality travel together before constructing the packet source calibration. Conversely, a source packet/product calibration together with the Kummer and mono-analytic forgetting transfers reconstructs this packet-local alignment object; the packet-local/(Ind3)-source equality is derived through the transfer chain rather than supplied as a parallel datum. This recovery has now been lifted to the realified-Frobenioid packet source: Lean uses the source object's own Kummer and mono-analytic forgetting compatibilities to reconstruct the packet-local source alignment. At the realified Step (x) entry level, the same projection also carries the upper-semi membership of the selected nonarchimedean entry. At the theta-root attached level, the projection also carries the canonical bad-local theta-root generator and nonzero full-label facts. The final Hodge/SHE/IPL/hull route now has an audit form that returns this recovered packet-local source alignment together with the C_Θ dichotomy. The target-only (Ind3) alignment is also derived at route level from the Step (x) packet-normalized theta source and the exact vertical-IQ target, with the $FMOD$-exactness guard retained. An audit form of this route exposes the derived equality between the Hodge-descent theta average and the (Ind3) target value. At the packet-correspondence layer, this source-alignment object and the exact vertical-IQ target now construct one log-Kummer correspondence source. The realified-Frobenioid packet source uses this correspondence-level constructor, so source and target Step (x) calibrations enter through a single packet source before the theta-root entry source is formed. The same correspondence-level reduction is now available for the packet-normalized ordered target: the named source-alignment object constructs the source face before the theta-average and target (Ind3) alignments construct the target face. The theta-root attached entry source now also projects this packet correspondence, so the canonical bad-local label facts and the full Step (x) transfer/target calibration endpoint are exposed from one source object. The Hodge/SHE/IPL/hull route has the same realified endpoint: its alignment record supplies the Hodge–Arakelov Gaussian comparison, canonical-one preservation, q-pilot positivity, and theta/Hodge log-volume calibration, while the Step (x) data enters through the single theta-root attached realified source. The route now also has a source-reduced endpoint: the theta-root label package, entry membership, packet-local source alignments, packet-normalized target alignment, and realified Frobenioid compatibility construct the Step (x) source inside the theorem before proving the same $C_\Theta = -1$ boundary alternative or strict $C_\Theta > -1$ conclusion. For the vertical-IQ form, the theorem now constructs the theta-root attached realified Step (x) source object directly from this source-alignment object and the exact IQ-target before applying the Hodge/SHE/IPL/hull endpoint. At the non-theta-root realified entry level, the corresponding source-aligned vertical-IQ endpoint now exposes the $FMOD$-exactness guard itself, together with Kummer/forgetting product preservation, target packet normalization, q-pilot non-interference, and the upper-semi inequality. The same exactness exposure has been lifted through the bad-local theta-root source: the canonical theta-root label data, exact vertical-IQ condition, packet-normalized target equalities, and Step (x) upper-semi inequality now appear in one theta-root attached endpoint. The packet-normalized theta-source calibration is now also consumed through a single Step (x) alignment object: the constant $ZMod(\ell)$ packet-normalized route and the proof that it has the same theta source as the Hodge-descent route are bundled before Lean derives the theta-average-to-packet-normalized equality used by the final corridor. The ordered-real-line form now has the same source-aligned boundary: the packet-local source-alignment object and realified Kummer/forgetting compatibility construct the source calibration internally before the matched theta-target alignment is used. This has been narrowed further to a target-only (Ind3) alignment: source-side packet-to-(Ind3) information is taken from the packet-local source-alignment object, while only the theta-average-to-target equality remains as ordered-real input. That target-only alignment is now constructible from Step (x) packet-normalized theta-source alignment together with an exact vertical-IQ target: Lean derives both sides through the common packet-normalized value and keeps the $FMOD$-exactness guard visible. The corresponding realified-source audit removes the independent packet-local source-alignment parameter: the alignment is recovered from the theta-root attached realified log-Kummer entry source, while the exact vertical-IQ target supplies the target-side equality and $FMOD$ guard. The stronger realified-source audit also removes the separate packet-normalized theta-source route at this boundary: theta packet normalization is projected from the same realified log-Kummer entry source, and the vertical-IQ target supplies only the (Ind3)-target packet-normalized equality. The Milestone 2 hull layer now contains a direct formalization of Remark 3.9.5(ii)–(iii)'s hull-map skeleton and approximant family. The hull map $\phi(P)$ is represented by a closure operator; Lean exposes the Remark 3.9.5(ii) laws

$$\phi(H) = H, \qquad P \subseteq \phi(P), \qquad P_1 \subseteq P_2 \Rightarrow \phi(P_1) \subseteq \phi(P_2),$$

together with idempotence $\phi(\phi(P)) = \phi(P)$. For a region P, an approximant record carries a hull H, proves $H \subseteq \phi(P)$, and proves

$$\mu_{\log}(P) \le \mu_{\log}(H) \le \mu_{\log}(\phi(P)).$$

For the Corollary 3.12 possible-image union, the public endpoints now also return $P \subseteq \phi(P)$, each $P_i \subseteq \phi(P)$, q-pilot containment in $\phi(P)$, and $\phi(\phi(P)) = \phi(P)$. The $\Xi(P)$-part of Remark 3.9.5(iii) is now represented at the same finite level: an exact hull approximant is a $\Phi(P)$-approximant H with $\mu_{\log}(H) = \mu_{\log}(P)$. When P is already closed under the hull operator, Lean constructs the canonical exact approximant. The possible-image Step (xi) source can now also be built from such an exact approximant and a weighted determinant source; in that route Lean derives $\mu_{\log}(P) = \mu_{\log}(H) = \det_{\log}$, rather than only the inequality $\mu_{\log}(P) \le \mu_{\log}(H)$. Remark 3.9.5(iv)'s statement $\phi(P) \in \Phi(P)$, hence $H_{\Phi(P)} = \phi(P)$, is now represented by a finite approximant-family object. Every member of the family is a $\Phi(P)$-approximant inside $\phi(P)$, and if one distinguished member is the canonical hull, Lean proves that the family union and its log-volume are exactly those of $\phi(P)$. The corresponding $\Xi(P)$-family statement is also represented. Each member is an exact approximant with log-volume $\mu_{\log}(P)$; if the family contains the canonical hull, its union is $\phi(P)$. In the special case where P is already closed under hull formation, the canonical one-point $\Xi(P)$-family has union $P = \phi(P)$ and log-volume $\mu_{\log}(P)$, matching Remark 3.9.5(iv)'s closed-region case. The canonical hull $\phi(P)$ is itself an approximant, and every approximant is fixed by hull formation. This is the compactness guard emphasized in the paper: the indeterminacy in choosing a hull approximant remains inside $\phi(P)$. The bounded-family version of Remark 3.9.5(v)–(vi) is now represented by a family-hull quotient source. For a family P_β, the code forms the hull of the union, uses it as the collapsed subset in the upper-semi quotient, and proves that any two nonempty possible regions have the same quotient image. The map statement in Remark 3.9.5(v) is now lifted to this bounded-family source: any set map from a nonempty possible region P_i into another possible region P_j induces, after quotienting by the family hull, the same image as the quotient map on P_i itself. The "formal empty set" extension in the same remark is represented by a finite formal-intersection system: a family of nonempty subsets of the collapsed subset S. Lean proves that every component has quotient image $\{S\}$, that all component images agree, and that component maps induce the same collapsed image after quotienting. Remark 3.9.5(vi)'s expected-equivalence statement is also represented for the bounded-family hull: for arbitrary nonempty regions A, B, their quotient images are both the collapsed point if and only if $A \subseteq \phi(P_B)$ and $B \subseteq \phi(P_B)$. The Ob5 compatibility of this quotient formalism with $\det^{\otimes M}(\phi(P_B))$ and $\mu_{\log}(\phi(P_B))$ is now represented at finite log-volume level: the same family hull that collapses the possible regions is identified with the weighted determinant log-volume, and the associated tensor-power normalization recovers this family-hull log-volume. Lean now also derives $\mu_{\log}(\bigcup_\beta P_\beta) \leq \det_{\log}$ from hull monotonicity and this determinant identification. The public Ob5 endpoint exposes these facts together with the expected-equivalence condition: for nonempty regions A, B, both quotient images are the collapsed family-hull point if and only if $A, B \subseteq \phi(P_B)$. The Ob9 log-volume passage is now represented at the same finite hull level. A realified semi-simplification correspondence between two hull shadows sends closed hulls to closed hulls, has a two-sided inverse on closed hulls, and preserves hull log-volume. Applied to the bounded-family hull, Lean proves that the transported hull still has log-volume $\det_{\log}$. This is the coarse log-volume content of Ob9; it is not yet a construction of the full log-link or its arbitrary-iterate indeterminacy. The Corollary 3.12 possible-image approximant source now projects the same estimate directly, so the upper-ray route can read $\mu_{\log}(P) \leq \det_{\log}$ from that source object. Remark 3.9.6 is represented by a finite product-formula theorem: if the log-link preserves the local log-volume at every place and the source and target conversion ratios agree, then the weighted global arithmetic degree sums agree. This has also been specialized to the Corollary 3.12 possible theta-image family, so the possible-image hull used by the upper-ray route and the Ob5 family-hull quotient/determinant source are definitionally the same hull. The canonical specialization now exposes this family-union-to-determinant estimate at its endpoint. The Corollary 3.12 possible-image approximant source projects to this family-hull quotient source, so the possible-image family used in the upper-ray route also carries the corresponding choice-independence theorem. Since the q-pilot region is contained in the chosen approximant and the approximant lies in the same family hull, the code also proves that a nonempty q-pilot region has the same family-hull quotient image as any nonempty possible theta-image choice. The same approximant now feeds the Step (xi) upper-ray constructor. If the q-pilot region lies in an approximant H, and if $\mu_{\log}(H)$ is the determinant-normalized log-volume, then Lean derives

$$\mu_{\log}(q) \leq \mu_{\log}(H) \leq \mu_{\log}(\phi(P)), \qquad \mu_{\log}(q) \leq \det_{\mathrm{norm}}.$$

Thus this part of the corridor no longer has to use only the canonical hull itself; it can use any certified hull-approximant inside $\phi(P)$. This has now been specialized to the possible-image union in Corollary 3.12. One source object carries the family of possible theta images, the union P, a certified approximant $H \in \Phi(P)$, $q \subseteq H$, and determinant normalization of $\mu_{\log}(H)$. The Step (x)-to-Step (xi) comparison is now expressed by a named handoff proposition over this fixed corridor and hull source. Thus the two numerical identifications $\mu_{\log}(H) = \mathrm{Ind3}_{\mathrm{upper}}$ and $\mu_{\log}(q) = \overline{\mu}_{\log}(q)_{\text{pre-Ind}}$ are no longer loose parameters of the endpoint; they are the explicit source obligation required before the Step (xi-f) upper-ray datum is formed. Lean then derives the whole chain

$$\mu_{\log}(P) \leq \mu_{\log}(H) \leq \mu_{\log}(\phi(P)), \qquad \mu_{\log}(q) \leq \mu_{\log}(H) \leq \det_{\mathrm{norm}}.$$

The determinant side has also been sharpened at finite log-volume level. Following Remark 3.9.5(vii), (Ob3-1)–(Ob3-3), a localization summand now carries a rank > 1, a bundle log-volume, a structure-sheaf adjustment, and a positive weight. A determinant source forms the finite weighted sum of the adjusted summands, then derives the positive tensor-power log-volume and the normalized determinant log-volume by definition. This replaces supplied tensor-power and normalized values for this new route, but it remains a log-volume skeleton rather than a construction of arithmetic vector bundles. The localization input itself has been refined one step further: a finite direct-summand source computes the bundle log-volume as a sum of direct summand log-volumes before projecting to the localization summand interface. Thus the weighted determinant can now be built from finite direct-summand localization sources. The Ob3-1-2 structure-sheaf adjustment is also named explicitly. A structure-sheaf adjusted source carries

$$L_{\mathrm{adj}} = \sum_{\gamma} L_\gamma - L_{\mathcal{O}(-)}$$

and proves that its weighted value is the same adjusted summand consumed by the determinant source. Thus the current determinant route separates direct summands, subtraction of the $\mathcal{O}(-)$ contribution, positive weighting, tensor-power formation, and normalization as distinct Lean interfaces. The possible-image approximant route can now consume this weighted determinant source directly. The equality between the approximant log-volume $\mu_{\log}(H)$ and the weighted normalized determinant value is now carried by a named Ob3-3 compatibility certificate. After this certificate is supplied, the old determinant record is only a projection used by the common upper-ray interface. At the package-ledger boundary, the hull/determinant source data can now be constructed from a concrete common hull K, a proof $P_{\mathrm{out}} \subseteq K$, and a determinant/log-volume estimate $\mu_{\log}(K) \leq \theta_{\mathrm{signed}}$. The resulting Lean endpoint returns target-union containment, the determinant bound, all-target bounds, and equality between the package's named hull/determinant bridge and this constructed bridge. Thus the remaining source obligation is the construction of K and its bound, not an opaque common-bound bridge. A more source-shaped constructor specializes K to $\phi(P_{\mathrm{out}})$ for an abstract hull operator ϕ. Its endpoint proves that the bridge hull is definitionally this hull of the target union, and reduces the source input to the determinant/log-volume estimate for $\phi(P_{\mathrm{out}})$ plus equality with the package's named bridge. The Set-based holomorphic-hull shadow now induces the Region-level hull operator and Region measure used by this package bridge when the underlying type is the point set of a real-line copy. The endpoint identifies the Region hull with the original point-set hull, proves region containment, and recovers monotonicity of the induced log-volume. At the package level, this induced hull operator can now be used directly: if the charted package measure is identified with the induced Region measure and the point-set hull of P_{out} satisfies the determinant bound, then the hull/determinant source data is constructed from the holomorphic-hull shadow. The remaining bridge equality is explicit, so the package cannot display one hull construction while proving with another. The source package's theta-pilot possible-image record now also feeds the canonical hull approximant directly: Lean proves that its set-theoretic union is exactly the package target union before applying the determinant comparison. The possible-image approximant source now feeds this package bridge in the canonical case. If its possible-image union is the package target union, its chosen approximant is the canonical hull, the package measure is the induced Region measure, and the normalized determinant value is at most θ_{signed}, then Lean constructs the same hull/determinant source data. No general containment of an arbitrary approximant is assumed. This has been lifted to the Theorem 3.11 source constructor: the approximant union may be tied to the source record's theta possible-image union, and the record then supplies the equality with the package target union. Thus the constructor no longer needs prebuilt hull/determinant source data in this canonical case. The same constructor now accepts the finite weighted determinant source from Remark 3.9.5(vii), (Ob3), directly. The determinant bound is stated for the weighted determinant log-volume, and Lean derives the normalized determinant bound consumed by the canonical hull bridge. A further specialization constructs the approximant as the canonical hull $\phi(P)$ itself. In this route the q-pilot input is containment in $\phi(P)$, and Lean derives the approximant-equals-hull proof from the canonical approximant constructor instead of accepting it as a separate input. Remark 3.9.5(vii), (Ob4), is represented at the same finite level: a naive Frobenius/tensor-power log-volume object records the passage $L \mapsto ML$ for $M > 0$, and Lean proves that normalized log-volume recovers L.

The weighted determinant source projects to this object, so the M-th tensor-power copy does not add a further log-volume ambiguity in the current corridor. A comparison object also proves that two positive tensor-power presentations with the same base log-volume have the same normalized log-volume, even when the tensor degrees differ. The package hull/determinant bridge now has an Ob4-shaped input form: a bound on the normalized tensor-power log-volume of the weighted determinant source is converted by Lean to the determinant bound used by the canonical hull bridge. In the canonical-hull form, this Ob4 input also derives the approximant-equals-$\phi(P)$ witness internally; the source input is q-pilot containment in $\phi(P)$, tensor-power determinant compatibility, and the bridge equality with the package hull/determinant datum. The same endpoint is exposed at the Theorem 3.11 source-record level, so the possible-image union used by the canonical hull is the record's multiradial theta-image union before being identified with the package target union. A record-native specialization removes the arbitrary possible-image-family input: the family is $i \mapsto P_i$ from the Theorem 3.11 source record, and Lean proves that its set-theoretic union is the record's theta-image union. Below the packet layer, the code now has a first realified Frobenioid degree source. A finite Frobenioid object carries a divisor degree, a unit log-volume, and their realified log-volume; a tensor-packet product may be declared to have product log-volume equal to this realified value. A finite divisor-monoid source now computes this divisor degree from prime multiplicities and prime degrees, matching the finite shadow of the divisor monoid in IUT I, Example 3.5. The same source now has a divisor-level tensor product that adds multiplicities and realified log-volumes when the prime-degree data agree; its projection to the degree-object layer agrees with the existing degree-object tensor product. It also has the Ob4 naive Frobenius tensor-power operation: multiplying the divisor multiplicities and unit log-volume by M constructs a new finite Frobenioid object whose divisor degree and realified log-volume are proved to be M times the original values. The older realified degree-object layer now has the same Ob4 operation, and the projection from the finite divisor-monoid source to that degree-object layer is proved to commute with tensor-power formation. The theta-version of IUT I, Example 3.5(ii), is now represented at this finite level: the same divisor degree is read through a formal $\log(\Theta)$-generator, and normalization of this generator to 1 recovers the ordinary finite divisor source. The Ob4 tensor-power operation also lifts to this theta copy, where Lean proves that the theta divisor log-volume is multiplied by M. The D-, or log-shell, version of Example 3.5(iii) is represented similarly: a distinguished local Frobenius/log-shell reading is divided by the extension degree before it is applied to the finite divisor degree. Tensor-powering this copy leaves the normalized Frobenius reading fixed and multiplies the resulting log-shell divisor log-volume by M. The three finite copies now have a compatibility package: if they share the same base divisor source and the theta/log-shell readings are normalized, then their log-volume readings are equal. This compatibility package is now stable under simultaneous Ob4 tensor-powering: the ordinary, theta, and log-shell copies are tensor-powered over the same object, the base identifications and normalizations are preserved, and all three finite log-volume readings are multiplied by M. The environment/Gaussian realified Frobenioid comparison of IUT II, Corollary 4.6(v), is represented at the same finite level by an evaluation package between two such compatible copy systems. The package stores only the ordinary divisor-source identification; Lean derives equality of the ordinary, theta, and log-shell log-volume readings across the environment and Gaussian sides from that identification and the copy-compatibility lemmas. The comparison is also stable under simultaneous Ob4 tensor-powering: the ordinary-source equality is preserved, the theta/log-shell comparison is rederived on the tensor-powered evaluation, and the two ordinary realified log-volume readings are multiplied by M. The same comparison is now connected to the finite local-global restriction layer, reflecting the F-prime-strip formulation following Corollary 4.6(v): if the environment and Gaussian global objects are the degree projections of their ordinary divisor sources and their local restrictions agree pointwise, then Lean derives equality of the global realified log-volumes and of all local restricted log-volume readings. The prime-index part of the same F-prime-strip statement is now explicit: finite prime-label types for C_{env} and C_{gau} come with bijections to the common valuation set V, and the evaluation map between prime labels is required to commute with these bijections. Lean then transports the local compatibility theorem along this prime-place square, deriving equality of restricted and local realified log-volumes at evaluated primes. The next functorial step toward the $\Theta^{\times\mu}$- and $\Theta^{\times\mu}_{\mathrm{gau}}$-links of Corollary 4.7(iii) is also represented at finite level. A lift to an $F^{\times\mu}$-prime-strip stores an environment-side unit character; the Gaussian-side unit character is defined by transport along the prime-evaluation equivalence. Thus Lean proves unit-character compatibility at evaluated primes rather than accepting it as an independent bridge field. The coric $F^{\vdash\times\mu}$ invariance of Corollary 4.7(iv) is represented by a common unit character on the valuation set V. Once the environment unit character is the pullback of this coric character along $Prime(C_{\mathrm{env}}) \simeq V$, Lean derives the corresponding Gaussian equality from the prime-evaluation square. The coric global realified Frobenioid compatibility of Corollary 4.7(v) is represented by a positive real scale on finite local-global Frobenioid collections. This is the finite shadow of the $R_{>0}$-orbit language: given scaled equality of the global realified log-volume and of the restricted local prime readings, Lean derives the scaled local-object and local-prime log-volume equalities from the existing local-global collection laws. The Frobenius-picture graph of Corollary 4.7(vi) is represented by the directed successor relation $n \to n+1$ on $\mathbb{Z}$. Lean proves that translations preserve this oriented graph and that any map exchanging the adjacent vertices 0 and 1 cannot preserve directed edges. This captures the finite order-theoretic content of the paper's statement that the chain has translation symmetry but not arbitrary permutation symmetry. Proposition 3.10(ii)'s column log-Kummer correspondence is now represented at the same finite level. For each vertical coordinate $m \in \mathbb{Z}$, the source carries a packet of group elements acting on one common IQ-carrier labelled by $n, \circ$. Translation in the column transports group elements, and Lean proves that the transported element acts by the same endomorphism on the common carrier. This records the translation-invariance of the log-Kummer action; it does not yet construct the underlying number fields. The realified-Frobenioid part of Proposition 3.10(iii) is represented at degree level by a family of realified Frobenioid objects indexed by $m \in \mathbb{Z}$. The log-link translation law preserves the extracted realified log-volume, including adjacent $m \mapsto m \pm 1$ shifts. The associated etale-picture core of Corollary 4.11(i),(ii) is represented by a common $D^{\vdash}$-core projection for all integer-labelled spokes. Lean proves that this projection is constant across successor steps and is unchanged by arbitrary permutations of the integer-labelled spokes, capturing the finite mono-analytic core/permutation-symmetry contrast with the Frobenius-picture chain. The value-group restriction discussed in Remark 4.10.3 is exposed as a combined finite-label endpoint: for a Gaussian degree evaluation, the canonical full label $j = 1$ has degree equal to the environment q-degree, the canonical square weight at $j = 1$ is 1, and the singleton restriction to $j = 1$ is not stable under translation by 1. Thus the identity/pivotal specialization and the failure of compatibility with the full label symmetry are checked in the same typed statement. A finite tensor-product operation on these degree objects adds divisor degrees and unit log-volumes, and Lean proves that the realified log-volume of the tensor product is the sum of the two realified log-volumes. This operation is also lifted to the tensor-packet source layer: if an existing tensor-packet product source is certified to carry the tensor-product Frobenioid degree, Lean derives additivity of its packet product log-volume. A Frobenioid Kummer-compatibility certificate then projects to the coric tensor-packet transfer by deriving equality of product log-volumes from equality of realified Frobenioid log-volumes, while retaining the Hodge-theater history guard. This is still a finite degree-level shadow, not a construction of Mochizuki's Frobenioids, but it moves one equality source below the packet-correspondence interface. The same degree layer now has finite Frobenioid morphisms. A morphism records the divisor-degree shift and unit-log-volume shift between two realified objects; Lean derives the corresponding shift in realified log-volume and proves identity and composition laws at this degree/log-volume level. This models the additive content of the pull-back and structure morphisms appearing in Remark 3.9.5(ix), (cQ3). These morphisms now transport across Ob4 tensor-powering: the source and target degree objects are tensor-powered, and the divisor and unit shifts of the induced morphism are multiplied by M. The Ob8/Ob9 vertical-shift issue is now attached directly to the Step (xi-g) handoff. A log-Kummer vertical-shift calibration pairs the Step (x)-to-hull upper-ray source with the global realified Frobenioid calibration. Lean proves that the calibrated value is the hull boundary used by the upper ray, that both q-pilot computations are bounded by this calibrated value, and that a shifted local value coincides with the calibrated value only when the shift term vanishes. Remark 3.9.5(ix)'s distinction between (cQ3),(cQ4) and (iQ5) is now represented by a finite closed-loop guard. The guard ties the hull/determinant log-volume obtained from the (sQ3) route and the calibrated log-Kummer log-volume obtained from the (sQ4) route to the same Step (xi) boundary. Lean then proves that the two values coincide, while the direct shifted local log-volume agrees with this boundary exactly in the zero-shift case. This does not construct $F^{\times\mu}$-prime-strips; it records the present formal boundary between the source-faithful route and the log-volume-only quotient. The global-to-local realified restriction from IUT I, Example 3.5 is also encoded at degree level: the restricted global prime log-volume is $[K_v : (F_{\text{mod}})_v]^{-1}$ times the local prime log-volume, and Lean proves that multiplying by the extension degree recovers the local prime log-volume. The same restricted value now normalizes the local p_v^N-shift source: the local shift is expressed with this restricted global prime step, and the automorphism-or-zero-step criterion is restated in terms of the restricted realified prime log-volume. Remark 3.9.5(ix)'s local-global collection language is represented at the same finite degree level. A collection contains a local realified Frobenioid object for each v, one global realified Frobenioid object, and localization restriction factors. Lean proves that each local log-volume is the restricted global value and that multiplying by the corresponding extension degree recovers the global log-volume. This is a finite log-volume shadow of the local-global category, not a construction of the category itself. This local-global collection now also backs the finite cQ3/cQ4 closed-loop guard: if the collection's global realified log-volume is the cQ3 hull/determinant boundary, then Lean proves that every localized value rescales to both the cQ3 boundary and the cQ4 calibrated log-Kummer boundary. Thus the Step (xi) boundary can be read from the local-global collection rather than from a bare real number alone. The local-global collection has also been refined by structure morphisms in the finite Frobenioid degree category. For each v, a degree morphism from the local object to the global object records divisor and unit shifts. Lean proves that the resulting morphism shift is exactly the additive form of the localization rescaling. This models the degree/log-volume content of the "structure poly-morphism" language in Remark 3.9.5(ix), (cQ3). The same finite category now has compatible morphisms between two such local-global structure collections. A compatible morphism consists of local morphisms at each v, one global morphism, and a commutativity condition: the local-then-structure path and the structure-then-global path have the same divisor and unit shifts. Lean derives equality of the total realified log-volume shift along these two paths. IUT I, Example 5.1 and Remark 5.1.1 distinguish the diagonal Frobenioid $F_{\langle J\rangle}$ from the product of underlying categories F_J: the former carries constant distributions in j, while the latter allows arbitrary distributions. The code now models this finite realified log-volume distinction. A diagonal distribution has total log-volume $|J|$ times its common value, with the realified restriction factor propagated through this sum; a distribution with two unequal coordinates is proved not to come from a diagonal constant distribution. IUT I, Example 5.4(vii) and Remark 5.4.2 are represented by a finite pivotal distribution object: a chosen pivotal index supplies the common diagonal log-volume, the summed interpretation gives total $|J|$ times this pivotal value, and the averaged interpretation recovers the pivotal value itself. IUT I, Definition 5.5 is now represented at finite log-volume level by an NF-bridge mode object: the bridge target is either the summed capsule log-volume or the normalized averaged capsule log-volume. For the same capsule distribution, Lean proves that the summed bridge target is $|J|$ times the averaged bridge target; for a pivotal distribution, these specialize to $|J|$ times the pivotal value and to the pivotal value itself. The adjacent Θ-bridge and ΘNF-gluing condition of Definition 5.5(iii) are formalized at the same finite level: the Θ side reads the common capsule total, and a compatibility record requires the NF- and Θ-bridge morphisms to use the same index bijection and the same capsule distribution. Lean then proves that a summed NF target equals the Θ-target, while an averaged NF target becomes the Θ-target after multiplication by $|J|$. Corollary 5.6(iii) is represented by a finite gluing torsor: once compatible NF and Θ bridge halves have been fixed, the gluing parameter is modeled by the additive F_ℓ-torsor on $\mathbb{Z}/\ell\mathbb{Z}$. Lean proves zero action, additivity of the action, existence and uniqueness of the translation between two gluing parameters, and preservation of the shared index-bijection and capsule data. This layer is now connected to the nonarchimedean packet correspondence: a realified-Frobenioid log-Kummer packet source projects the Frobenioid compatibility data to the Kummer $F \to D$ transfer and to the mono-analytic forgetting transfer, and then feeds the existing packet correspondence endpoint. Thus the Step (x) source equality used by the route can be obtained from realified Frobenioid log-volume compatibility plus the remaining packet source/target calibrations. IUT III, Remark 3.11.4 further says that the root-of-unity and cyclotomic indeterminacies arising from forgetting Frobenius-like labels are invisible for the objects under consideration. The present code formalizes the finite sign-unit part of this assertion: a unit in the subgroup $\{1, -1\} \subset (\mathbb{Z}/\ell\mathbb{Z})^\times$ preserves the zero/nonzero full cusp label, every full-label log-volume expression, and the averaged Gaussian log-volume. This is not yet a construction of the full cyclotomic rigidity isomorphism; it is the finite-label component that the current Stage 1 interface can prove without an additional Frobenioid/cyclotome layer. The same finite-label layer now also proves the corresponding exclusion: an affine coordinate change $j \mapsto t + aj$ preserves the full cusp label if and only if $t = 0$ and a is represented by such a sign-unit invisibility certificate. For nonzero Gaussian environment degree, the same classification is equivalent to pointwise preservation of the Gaussian degree profile. Thus the current code has a finite analogue of the "no further indeterminacies" assertion of Remark 3.11.4(ii). The integrated route now composes the Hodge–Arakelov theta-value sources with this packet-target nonarchimedean source. The resulting theorem derives the same boundary/strict C_Θ dichotomy from source-facing Gaussian evaluation data and source-facing upper-semi data, rather than from two unrelated collections of raw equalities. For the target side, the strongest route now uses a target-only Step (x) alignment

$$\Theta_{\mathrm{avg}} = U_{\mathrm{Ind3}}$$

together with calibration of the local entry target against U_{Ind3}. The direct equality $\Theta_{\mathrm{avg}} = \mathrm{entry}_{\mathrm{target}}$ is then derived. When the theta-source average is identified with the packet-normalized capsule log-volume, this target alignment is itself derived from calibration of U_{Ind3} with the same packet-normalized value. The local entry target may now also be calibrated against this packet-normalized value, so the equality $\mathrm{entry}_{\mathrm{target}} = U_{\mathrm{Ind3}}$ is derived rather than supplied directly in the strongest route.

5.4 Step (xi-f) algebra

The final real algebra is fully formalized. In informal notation, the main theorem is:

Theorem 1 (Formal Step (xi-f))**.** *Assume $q < 0$, $q \leq \theta$, and $\theta \leq C(-q)$. Then $C \geq -1$. If $q < \theta$, then $C > -1$. If $C = -1$, then $q = \theta$ and $\theta < 0$.*

This theorem is instantiated by the Step (x)-to-Step (xi) route whenever the hull/log-volume upper-ray comparison and the displayed C_Θ-bound are available. It is also instantiated by the current nonarchimedean (Ind3)-plus-factored-SHE route, subject to the same positivity and displayed-bound hypotheses. Thus the remaining gap is not the final ordered-real implication, including its strict and boundary branches, but the construction of the source hypotheses consumed by that implication.

6 Boundary Quotient Experiment

The most developed diagnostic concerns Step (xii), where local Frobenioid p_v^N-ambiguity is compared with the globally calibrated log-volume. The formal model records a local shift

$$L_N = L_0 + N\delta, \qquad N \in \mathbb{Z},\ \delta \in \mathbb{R},$$

and a global calibration that fixes exponent 0. Lean proves

$$L_N = L_0 \iff N\delta = 0,$$

and, if $\delta \neq 0$,

$$L_N = L_M \iff N = M.$$

At the boundary branch $C_\Theta = -1$, the formal corridor identifies

$$q_{\text{pilot}} = \Theta_{\text{hull}} = L_0.$$

The local shift model is now source-facing: a local Frobenioid log-shell submodule carries a p_v^N-action whose log-volume effect is $L_0 \mapsto L_0 + N\delta$. The local action records whether the p_v^N-endomorphism is an automorphism, and in the current model this is equivalent to $N = 0$. The global realified Frobenioid calibration package selects exponent 0, hence its calibrated value is L_0. If the local endomorphism is not an automorphism and $\delta \neq 0$, Lean proves that the local shifted value cannot equal the globally calibrated value. For the actual upper-ray quotient that collapses $\{x \mid x \leq \Theta_{\text{hull}}\}$, Lean proves the exact classification

$$[L_N] = [q_{\text{pilot}}] \iff N\delta \leq 0,$$

and the complementary separation criterion

$$[L_N] \neq [q_{\text{pilot}}] \iff N\delta > 0.$$

If $\delta > 0$, this becomes:

$$[L_N] = [q_{\text{pilot}}] \iff N \leq 0, \qquad [L_N] \neq [q_{\text{pilot}}] \iff 0 < N.$$

This is not a proof for either side of the Mochizuki–Scholze–Stix dispute. It is a precise reading of one simplified quotient mechanism inside the current formal corridor. It shows that quotient collapse can identify unequal underlying log-volumes if they lie in the collapsed upper ray, while positive local shifts remain separated. The decisive question is therefore not the ordered-real algebra; it is whether the source IUT constructions actually supply the required calibrated hull/log-volume comparison without making an illicit identification of histories or comparison levels.

7 Current Results

The project currently proves the following source-facing results in Lean:

1. The finite q^{j^2} degree rule, sign quotient, half-range representatives, zero/nonzero split, and average formulas are formalized.

2. The bad-local theta-root quotient-$\mathbb{Z}$ package now feeds the Stage 1 $|F_\ell|$ full-label model through its canonical coordinate; the concrete $Z/\ell Z$ case yields the canonical nonzero full label.

3. A single label-independent scalar cannot absorb all representative j^2-degrees when $\deg(q) \neq 0$.

4. The raw representative branch does not descend to the sign quotient, while the balanced full-label Gaussian branch does.

5. The Step (x) averaged-log-volume corridor feeds a Step (xi) upper-ray comparison once determinant/hull identifications are supplied.

6. A nonarchimedean (Ind3) entry with factored square/full-label preservation now directly supplies the final weighted-theta route and the signed $C_\Theta \geq -1$ endpoint, once q-positivity and the displayed C_Θ-bound are supplied.

7. The same nonarchimedean route now exposes the strict branch $C_\Theta > -1$ and the boundary consequence $C_\Theta = -1 \Rightarrow q_{\text{signed}} = \Theta_{\text{signed}} < 0$. It also packages these as one dichotomy at the route level.

8. Gaussian degree evaluations with preserved environment degree now feed directly into this nonarchimedean C_Θ dichotomy, constructing the factored square/full-label SHE package internally.

9. In the identity-coordinate Gaussian route, equality at the canonical full label 1 supplies the environment-degree preservation needed by the same nonarchimedean C_Θ dichotomy.

10. The Hodge–Arakelov theta-value evaluation source now constructs the Gaussian degree evaluation from a theta-monoid degree and the bad-local theta-root label source; canonical-label preservation feeds the factored SHE square/full-label obligations.

11. A nonarchimedean upper-semi object equipped with 1-column q-pilot log-Kummer data now supplies the raw nonarchimedean (Ind3) entry alignment used by the canonical Gaussian route.

12. The source-side equality in that compatibility can be derived from product-log-volume preservation along the finite Kummer-plus-forgetting tensor-packet chain.

13. A theta-root attached Kummer/forgetting source package now combines the canonical bad-local $|F_\ell|$-label source with a source-side packet/product calibration object and that finite chain, deriving the nonarchimedean source equality.

14. Source-side and target-side packet calibrations are now joined by one nonarchimedean log-Kummer packet-correspondence source carrying the q-pilot non-interference guard. Its projections derive the theta-to-packet, theta-to-entry, entry-to-(Ind3)-target, and packet-normalized target equalities used by the route; the target calibration now refers directly to the packet-normalized capsule value rather than to a separate real representative.

15. The strongest Gaussian first-pass route can now consume this single packet-correspondence source and project both its source and target faces, instead of taking source packet equalities and target calibration as separate route inputs.

16. A realified-Frobenioid log-Kummer entry source now bundles the selected upper-semi inclusion with the realified packet source; the strongest Step (x) route projects Kummer transfer, mono-analytic forgetting, source/target packet calibration, and q-pilot non-interference from this single source.

17. A theta-root attached realified entry source now combines the canonical bad-local theta-root label package with that realified Step (x) source, and the Hodge–Arakelov route can consume this attached source directly.

18. The Hodge/SHE/IPL/hull route now also has a realified endpoint: route alignment supplies the Hodge–Arakelov and q-pilot data, and Step (x) enters through the theta-root attached realified source.

19. Remark 3.9.5(iii)'s hull-approximant family is now represented directly: an approximant lies inside the canonical hull $\phi(P)$, has log-volume between P and $\phi(P)$, and is fixed by hull formation.

20. The bounded-family upper-semi quotient of Remark 3.9.5(v)–(vi) is represented: each possible region lies in the hull of the family union, and any two nonempty possible regions have identical images in the quotient that collapses this family hull.

21. The Ob5 determinant/log-volume compatibility of this family quotient is represented at finite level: the same $\phi(P_B)$ quotient source is paired with a weighted determinant certificate, and Lean proves that the determinant log-volume and normalized tensor-power log-volume are the family-hull log-volume.

22. This Ob5 compatibility is specialized to the Corollary 3.12 possible theta-image family: the upper-ray source hull and the family quotient determinant/log-volume source share the same $\phi(P_B)$.

23. The canonical possible-image Ob5 endpoint now returns the family-union-to-determinant estimate, not only the quotient collapse and determinant equality.

24. The Corollary 3.12 possible-image approximant source now projects to this bounded-family quotient source, yielding the same quotient-image equality for nonempty possible theta-image choices.

25. In that same source, q-pilot containment in the chosen approximant implies q-pilot containment in the family hull; if the q-pilot region is nonempty, its quotient image agrees with the quotient image of any nonempty possible theta-image choice.

26. Such a certified hull-approximant now also constructs the Step (xi) upper-ray datum: if the q-pilot region is contained in the approximant and the approximant log-volume is determinant-normalized, then Lean derives the q-pilot bound by the approximant, by the canonical hull, and by the determinant log-volume.

27. The approximant route is now specialized to the Corollary 3.12 possible-image union. A single source object binds the possible theta-image family, its union, the chosen hull-approximant,

q-pilot containment in that approximant, and determinant normalization; the experiment endpoint returns the union/approximant/hull and q-pilot/determinant inequalities.

28. The determinant input can now be produced from a finite weighted localization source: adjusted vector-bundle summand log-volumes are summed, the positive tensor-power value is derived, and normalization is proved to return the determinant log-volume before projecting to the determinant interface.

29. Each localization summand can now itself be produced from a finite direct-summand source: the bundle log-volume is the sum of direct-summand log-volumes, then the structure-sheaf adjustment and positive weight produce the adjusted localization contribution.

30. The Ob3-1-2 structure-sheaf adjusted source makes the $\mathcal{O}(-)$ subtraction explicit before projection to the localization summand. A corresponding determinant constructor proves that summing these weighted adjusted sources gives the determinant log-volume used by the route.

31. The possible-image hull-approximant endpoint now has a weighted determinant-source constructor. The route consumes the weighted summand source and projects to the common upper-ray interface only after a named Ob3-3 compatibility certificate identifies the approximant log-volume with the weighted normalized determinant value.

32. The weighted determinant route now has a canonical-hull specialization: the approximant is constructed as $\phi(P)$, q-pilot containment is stated directly in this hull, and the approximant-equals-hull proof is derived by Lean rather than supplied as a separate route input.

33. The Ob4 tensor-power passage is now represented at log-volume level: M-th tensor-power formation multiplies by $M > 0$, and normalization recovers the base log-volume. The weighted determinant source projects to this tensor-power object, and two tensor-power presentations with the same base log-volume have equal normalized log-volume.

34. The bounded-family hull/determinant source now proves the direct estimate from possible-image union to determinant log-volume: $\mu_{\log}(\bigcup_\beta P_\beta) \leq \det_{\log}$.

35. The Corollary 3.12 possible-image approximant source now exposes the same possible-image-union-to-determinant estimate directly.

36. The package-level hull/determinant bridge can now be built from the Ob4-normalized tensor-power bound associated to a weighted determinant source. The conversion to the determinant bound is a Lean theorem, not a separate bridge assumption.

37. A first realified Frobenioid degree layer is present: equality of realified Frobenioid log-volumes now derives the product-log-volume equality used by the tensor-packet coric transfer, with theater histories still kept distinct. The degree object also has a finite tensor-product operation: divisor degree, unit log-volume, and realified log-volume are proved to add. A tensor-packet source certified by this Frobenioid product inherits product-log-volume additivity.

38. The realified Frobenioid degree object can now be constructed from a finite divisor-monoid source: Lean computes the divisor degree as $\sum_p m_p \deg(p)$ before adding the unit log-volume.

39. The finite divisor-monoid source now has its own tensor product: multiplicities and unit log-volumes add, and Lean proves additivity of the induced divisor degree and realified log-volume.

40. The same finite divisor-monoid source now has an Ob4-style naive Frobenius tensor-power operation: multiplicities and unit log-volume are multiplied by M, and Lean proves the induced divisor degree and realified log-volume are multiplied by M.

41. The older realified degree-object layer now has the same Ob4 tensor-power operation, and the projection from a finite divisor-monoid source to this layer commutes with tensor-power formation.

42. The projection of this divisor-level tensor product to the realified degree-object layer agrees with the existing degree-object tensor product, preventing these two Frobenioid shadows from drifting apart.

43. The theta-version of the finite divisor source is now represented: a formal $\log(\Theta)$-generator scales the divisor degree, normalization to 1 recovers the ordinary source, and tensor products preserve additivity when the theta generators match. The Ob4 tensor-power operation also lifts to this copy, multiplying the theta divisor log-volume by M.

44. The D/log-shell version is also represented at finite level: Lean normalizes a distinguished local Frobenius log-volume by the extension degree, recovers the ordinary source when this normalized value is 1, and proves tensor-product additivity when normalized values match. The Ob4 tensor-power operation preserves the normalized Frobenius reading and multiplies the log-shell divisor log-volume by M.

45. The ordinary, theta, and log-shell divisor copies now have a compatibility package: shared base divisor data plus normalized readings imply equality of all three finite log-volume readings. Simultaneous Ob4 tensor-powering of the three copies preserves this compatibility package and multiplies all three finite log-volume readings by M.

46. The finite environment/Gaussian realified Frobenioid comparison attached to IUT II, Corollary 4.6(v), is now encoded as an evaluation package between two compatible copy systems. The only stored bridge is equality of the ordinary divisor source; Lean derives equality of the ordinary, theta, and log-shell readings across the two sides. Simultaneous Ob4 tensor-powering preserves the stored ordinary-source equality, rederives the theta/log-shell comparison, and multiplies the two ordinary realified log-volume readings by M.

47. This evaluation package is now tied to the existing finite local-global restriction layer: if the two global objects are the degree projections of the ordinary environment/Gaussian divisor sources and the local restrictions agree at every v, then Lean derives equality of the global realified log-volume and of the pointwise local restricted log-volume readings.

48. The prime-index skeleton of the corresponding F-prime-strip evaluation is now explicit: finite prime-label types for C_{env} and C_{gau} carry bijections to V, and the evaluation equivalence between prime labels must commute with these bijections. Lean transports the local compatibility theorem along this square to obtain equality at evaluated primes.

49. The finite $F^{\times\mu}$-prime-strip lift used by the $\Theta^{\times\mu}$- and $\Theta^{\times\mu}_{\mathrm{gau}}$-links is now represented by transport of a unit character along the prime-evaluation equivalence. The Gaussian unit character is constructed from the environment unit character, and Lean proves compatibility at evaluated primes.

50. The coric $F^{\vdash \times \mu}$ invariant of Corollary 4.7(iv) is now represented at finite level: a common unit character on V pulls back to the environment side, and Lean derives the Gaussian-side pullback equality from the prime-evaluation square.

51. The coric global realified Frobenioid compatibility of Corollary 4.7(v) is now represented by a positive real scale on finite local-global Frobenioid collections. From scaled global and restricted-local readings, Lean derives the scaled local-object and local-prime log-volume readings.

52. The Frobenius-picture graph of Corollary 4.7(vi) is now represented as the directed successor graph on $\mathbb{Z}$. Lean proves translation invariance and rules out edge preservation by any map that swaps adjacent vertices 0 and 1.

53. The etale-picture $D^{\vdash}$-core of Corollary 4.11(i),(ii) is now represented as a common core projection for all integer-labelled spokes. Lean proves successor-step constancy and invariance under arbitrary permutations of the spoke labels.

54. The value-group restriction of Remark 4.10.3 is now a single endpoint: the Gaussian degree at $j = 1$ equals the environment q-degree, the square weight at $j = 1$ is 1, and the singleton-1 restriction is not translation-invariant.

55. The realified Frobenioid degree layer now has finite morphisms: divisor and unit shifts determine the target realified log-volume, identity has zero shift, and composition adds both shifts. These morphisms are stable under Ob4 tensor-powering, with divisor and unit shifts multiplied by M.

56. The Ob8/Ob9 vertical-shift guard is now tied to the Step (xi-g) two-computation handoff: global realified Frobenioid calibration selects the hull upper-ray boundary, both q-pilot readings are bounded by it, and shifted local coincidence is equivalent to a zero shift term.

57. The cQ3/cQ4 portion of Remark 3.9.5(ix) now has a finite closed-loop guard: the hull/determinant value and the calibrated log-Kummer value are proved equal as the same Step (xi) boundary, while the direct log-volume-only shifted value agrees with that boundary only under the zero-shift condition.

58. The global-to-local realified Frobenioid restriction factor of IUT I, Example 3.5 is formalized as inverse-extension-degree scaling of the local prime log-volume, with the degree-normalization identity proved in Lean. This restricted value is connected to the local p_v^N-shift source used in the Step (xii) diagnostic.

59. A finite local-global realified Frobenioid collection now models the degree/log-volume content of Remark 3.9.5(ix): each local object is a restricted global value, and extension-degree rescaling recovers the global realified log-volume.

60. The cQ3/cQ4 closed-loop guard can now be backed by such a local-global collection: the global collection value is identified with the cQ3 hull/determinant boundary, and every localized value rescales to both the cQ3 and cQ4 Step (xi) boundaries.

61. The local-global collection now carries finite structure morphisms from local objects to the global determinant object. The morphism divisor/unit shift is proved equal to the additive localization rescaling term.

62. Compatible morphisms between local-global structure collections are now represented: local-then-structure and structure-then-global paths have equal divisor/unit shifts and hence equal total realified log-volume shift.

63. The diagonal/arbitrary distribution distinction of IUT I, Example 5.1 and Remark 5.1.1 is formalized at finite realified log-volume level: diagonal distributions are constant in j, their total is $|J|$ times the common restricted value, and a two-coordinate mismatch cannot be represented as diagonal.

64. The pivotal distribution of IUT I, Example 5.4(vii), is modeled by a chosen index in a diagonal distribution; Lean proves the summed log-volume formula and the normalized averaged formula of Remark 5.4.2.

65. The finite NF-bridge log-volume model of IUT I, Definition 5.5, separates summed and averaged bridge targets and proves their $|J|$ scaling relation for a common capsule distribution.

66. The finite ΘNF-bridge compatibility model records the same-index-bijection and same-capsule requirements of Definition 5.5(iii), and proves the corresponding Θ-target equalities for summed and averaged NF modes.

67. The finite gluing set of Corollary 5.6(iii) is modeled as the additive F_ℓ-torsor on $\mathbb{Z}/\ell\mathbb{Z}$, with compatibility data preserved under gluing translation.

68. The realified Frobenioid layer now feeds the nonarchimedean log-Kummer packet source: Kummer and mono-analytic forgetting transfers are projected from Frobenioid compatibility before the packet correspondence endpoint is applied.

69. A packet-normalized Step (x) entry-target source now constructs the target side of a nonarchimedean upper-semi entry from the capsule-family normalized log-volume. The target-calibration constructor derived from this source removes the separate entry-target-to-packet-normalized equality.

70. The same target-calibration layer can now derive the (Ind3)-target-to-packet-normalized equality from theta-target alignment and theta-average packet normalization, instead of accepting that equality directly.

71. A vertical-IQ target source now models the exact common-target side of the log-Kummer correspondence in Proposition 3.10(ii). It derives the theta-average, (Ind3)-target, and entry-target packet-normalized equalities used by the packet target calibration.

72. The vertical-IQ source carries an $FMOD$-exactness guard, reflecting Remark 3.10.1's distinction between exact $FMOD$ Frobenioid compatibility and one-sided $Fmod$ upper semi-compatibility.

73. The source-aligned realified Step (x) entry endpoint now exposes this $FMOD$-exactness guard at the same level as the Kummer/forgetting transfer chain, target packet-normalized equalities, q-pilot non-interference, and upper-semi inequality.

74. The theta-root attached source-aligned Step (x) endpoint now carries the same $FMOD$-exactness guard together with the canonical bad-local theta-root label facts.

75. The log-Kummer packet-correspondence source can now be constructed from source packet/product calibration plus the packet-normalized entry target and theta-target alignment. The q-pilot non-interference, source equality, and target equality endpoint is then recovered from this constructed correspondence.

76. The packet-correspondence endpoint now exposes the full finite Step (x) transfer chain: Kummer product preservation, mono-analytic forgetting product preservation, source equality with the (Ind3) real source, target packet-normalized equalities, and q-pilot non-interference.

77. The source packet/product calibration itself now has a constructor from packet-local/source alignment, packet-local (Ind3)-source alignment, and Kummer/forgetting product-log-volume preservation. The source holomorphic-product equality is derived by this constructor.

78. A calibrated source packet/product face can now be projected back to the packet-local source-alignment object once the Kummer and mono-analytic forgetting transfers are supplied; the packet-local (Ind3)-source equality is reconstructed through that transfer chain.

79. The same projection is now available from a realified-Frobenioid log-Kummer packet source itself: its internal Kummer and forgetting compatibilities supply the transfer chain needed to recover the packet-local source-alignment object.

80. The realified Step (x) entry source now projects the same packet-local source alignment while retaining the selected entry's upper-semi membership.

81. The theta-root attached realified Step (x) entry source now projects the same packet-local source alignment while retaining the canonical generator and nonzero full-label facts.

82. The Hodge/SHE/IPL/hull route has a source-audit endpoint: from one theta-root attached realified Step (x) source, Lean returns the recovered packet-local alignment equalities together with the C_Θ dichotomy.

83. The source-aligned Hodge/SHE/IPL/hull route now derives its target-only (Ind3) alignment from the Step (x) packet-normalized theta source and the exact vertical-IQ target source, retaining the $FMOD$-exactness guard at the route boundary.

84. The audit form of this vertical-IQ route also exposes the derived equality from the Hodge-descent theta average to the (Ind3) target value, before returning the C_Θ dichotomy.

85. The packet-correspondence constructor has a stronger alignment form: packet-local source alignment, packet-local (Ind3)-source alignment, packet-normalized entry target, and theta-target alignment generate both packet calibrations before projecting the q-pilot endpoint.

86. The realified-Frobenioid log-Kummer packet source has the same reduced target-data constructor: Frobenioid compatibility yields the Kummer and forgetting transfers, while packet-normalized target data is generated from the entry-target source and theta-target alignment.

87. The vertical-IQ target source has been lifted through the realified-Frobenioid entry source and the theta-root attached entry source, so their upper-semi endpoints no longer need independent target alignment equalities when common-target log-Kummer data is available.

88. The integrated Hodge/SHE/IPL/hull C_Θ-dichotomy route now has a vertical-IQ form: exact common-target log-Kummer data supplies the target equalities used by Step (x) before the route enters the final Corollary 3.12 real inequality.

89. The integrated vertical-IQ route also has a source-reduced form: source packet calibration is constructed internally from packet-local alignments and realified Kummer/forgetting compatibility.

90. The same source-reduced vertical-IQ data now constructs the realified Step (x) log-Kummer entry source itself, before any final C_Θ-dichotomy theorem is invoked.

91. The realified packet source now also has a source-reduced constructor: packet-local/source alignment and packet-local (Ind3)-source alignment generate the source packet calibration using the Frobenioid-derived Kummer and forgetting transfers.

92. The realified Step (x) entry source has the same reduced boundary: entry membership plus source/target alignment data constructs the packet source internally and projects q-pilot non-interference, source equality, target equality, and the upper-semi inequality.

93. The same realified packet source now exposes a combined calibration endpoint: it proves Kummer and forgetting product-log-volume preservation, packet-local equality with the entry source, entry-source equality with the (Ind3) source, packet-normalized equality for the theta average, (Ind3) target, and entry target, and q-pilot non-interference.

94. The theta-root attached realified entry source now has the same source-reduced vertical-IQ constructor: the upper-semi packet entry, packet-local source alignments, realified Kummer/forgetting compatibility, and exact vertical-IQ target construct one source object that projects the canonical nonzero theta-root label facts and the realified Step (x) upper-semi endpoint.

95. The packet-correspondence source now has the same source-aligned vertical-IQ constructor: a named packet-local source-alignment object and an exact vertical-IQ target source construct one correspondence object before the realified-Frobenioid layer is applied.

96. The packet-correspondence source also has a source-aligned packet-normalized ordered-target constructor: the source face is generated from the named packet-local alignment object, while the target face is generated from the packet-normalized entry and theta-target alignment.

97. The theta-root attached realified entry source now projects the underlying packet correspondence endpoint, combining canonical bad-local label facts with transfer-chain equalities, packet target calibrations, q-pilot non-interference, and the upper-semi bound.

98. The Hodge/SHE/IPL/hull route now consumes this reduced theta-root Step (x) data directly: the theorem constructs the realified source internally from a named packet-local source-alignment object and exact vertical-IQ target, then concludes the current C_Θ dichotomy.

99. The same route has an ordered-real-line form: the packet-source and theta-target (Ind3) real equalities are extracted from one matched-scale ordered transport alignment, so they are no longer accepted as unrelated route inputs at this boundary.

100. The ordered packet-normalized route now has a source-aligned form: packet-local source alignment plus realified Kummer/forgetting compatibility construct the source calibration internally before the route enters the theta-root attached Step (x) source.

101. The same source-aligned ordered route now has a target-only alignment form: the source-side (Ind3) equality is supplied by the source alignment object, and the route consumes only the theta-target (Ind3) equality as ordered-real input.

102. The target-only (Ind3) alignment can now be constructed from a Step (x) packet-normalized theta-source alignment and an exact vertical-IQ target source.

103. The realified-source vertical-IQ audit recovers the packet-local source-alignment object from the theta-root attached realified log-Kummer entry source, so this route boundary no longer accepts that source alignment as an independent input.

104. The stronger realified-source vertical-IQ audit also projects theta-average packet normalization from that same source, removing the separate Step (x) packet-normalized theta-source route from the audit boundary.

105. In the ordered form, the packet-normalized theta calibration can now be obtained from a constant $ZMod(\ell)$ packet-normalized route. The theorem requires equality of the Hodge-descent theta source with the packet-normalized theta source, preventing a packet-normalization audit for one theta source from being used for another.

106. The same source equality is now bundled with the packet-normalized route as a Step (x) theta-source alignment, and the Hodge/SHE/IPL/hull endpoint consumes that aligned object instead of separate route/equality inputs.

107. The selected packet-normalized nonarchimedean entry is now bundled with its membership in the audited upper-semi inclusion family. The route consumes this single source object rather than a constructed entry and separate membership proof.

108. The source side of that Step (x) entry can now be supplied by a named packet/product source calibration. The route no longer receives the packet-local/source and source/mono-analytic-product equalities separately; they are projected from the calibration object before the realified Frobenioid source is constructed.

109. The finite sign-unit/root-of-unity part of Remark 3.11.4 is formalized: sign-unit action preserves full cusp labels, full-label log-volume readings, Gaussian degrees, and averaged Gaussian log-volume.

110. The same finite layer classifies the allowed unit-affine label changes: full-label preservation is equivalent to zero translation plus a sign-unit invisibility certificate; under nonzero Gaussian environment degree this is also equivalent to pointwise Gaussian-degree preservation.

111. The current integrated route composes Hodge–Arakelov theta-value sources with the packet-target nonarchimedean source to obtain the same $C_\Theta = -1$ boundary alternative or strict $C_\Theta > -1$ conclusion.

112. A Theorem 3.11 multiradial source record now packages structured IPL/SHE/APT, coric output, possible theta images, pilot-column asymmetry, distinct log-shell treatments, and the no-history-identification guard.

113. The IPL source layer now has an explicit log-volume transport record with source and target Hodge-theater sides, prime-strip endpoint compatibility, log-volume preservation, and no-history-identification.

114. The SHE source layer now derives the current square/full-label preservation package from Hodge–Arakelov canonical-one synchronization, exposing pointwise representative comparison, transported-average equality, and the retained Hodge-history separation.

115. The Step (xi) hull/determinant layer now has a Theorem 3.11 source constructor: source obligations are derived from the source record where possible, the record's theta possible-image union is tied to the hull endpoint, and the determinant bound yields $q_{\mathrm{signed}} \le \theta_{\mathrm{signed}}$.

116. The theta-pilot possible-image family of the source package now feeds the canonical holomorphic-hull approximant directly: Lean identifies its set-theoretic union with the package target union before the determinant comparison is applied.

117. The package-level hull/determinant source bridge can now be constructed from a concrete common hull containing the target union and a determinant/log-volume bound for that hull. The experiment endpoint returns target-union containment, the determinant bound, all-target bounds, and equality with the package's named hull/determinant bridge.

118. The same source bridge has a hull-operator specialization: the common hull is $\phi(P_{\mathrm{out}})$ for the target-union region P_{out}, and the endpoint records this equality before deriving containment, determinant, and all-target bounds.

119. The Set-based holomorphic-hull/log-volume shadow now induces a Region-level hull operator and Region measure on real-line points. The experiment endpoint proves exact agreement with the original point-set hull, containment, and induced log-volume monotonicity.

120. The package-level hull/determinant bridge can now be constructed from that holomorphic-hull shadow when the package measure is identified with the induced Region measure and the point-set hull satisfies the determinant bound.

121. The possible-image approximant source can now construct the package hull/determinant bridge in the canonical case: the possible-image union must be the package target union, the approximant must be the canonical hull, and the normalized determinant value must be bounded by θ_{signed}.

122. The Theorem 3.11 hull/determinant constructor can now be built directly from the source record plus this canonical approximant data, with the record's possible-image union supplying the bridge to the package target union.

123. The same Theorem 3.11 constructor can now be built from a canonical possible-image approximant backed by the finite weighted determinant source of Remark 3.9.5(vii), (Ob3). The endpoint exposes the equality between the approximant log-volume and the weighted determinant log-volume.

124. The canonical weighted determinant constructor now has the stronger canonical-hull input form: the approximant is definitionally $\phi(P)$, q-pilot containment is imposed in this hull, and the endpoint exposes the derived approximant-region equality.

125. The canonical-hull route now has the same Ob4 tensor-power input form: a normalized tensor-power bound is converted to the determinant bound, while Lean derives the approximant-equals-$\phi(P)$ witness from the canonical hull constructor.

126. This canonical-hull Ob4 endpoint is also public for the Theorem 3.11 source constructor: the source record's theta possible-image union is the hull input before the package target-union identification is applied.

127. A record-native specialization now derives that theta-image union from the Theorem 3.11 record's own possible-image family, instead of accepting an arbitrary image family plus a separate union equality.

128. The Step (x)-to-Step (xi) upper-ray bridge can now be built directly from a certified possible-image hull approximant. The determinant normalization is read from the approximant/determinant compatibility, rather than supplied as a separate equality at the bridge boundary.

129. The Step (x)-to-Step (xi) hull comparison is now a named handoff proposition over a fixed procession corridor and fixed hull source; from this handoff Lean derives q-pilot upper-ray membership and determinant normalization at the Step (xi-f) boundary.

130. The same upper-ray bridge now has a log-Kummer vertical-shift calibration endpoint: the globally calibrated Frobenioid log-volume is the hull boundary used in Step (xi-g), not an independently chosen real value.

131. The Step (xi) endpoint now exposes the finite cQ3/cQ4 closed loop: hull/determinant passage and log-Kummer calibration produce the same boundary, and the shifted local log-volume route is equivalent to a zero-shift specialization.

132. A Hodge/SHE/IPL/hull packet-target route now derives the target-side log-volume equality from SHE synchronization and reads q-positivity from the hull constructor; the source-side theta/Hodge equality is derived from a full-label calibration certificate in the route-alignment source record.

133. The target-side theta-to-entry equality is no longer a direct input in the strongest route. It is derived from a Step (x) theta-target alignment and calibration of the entry target with the upper-semi target value.

134. The Step (x) theta-target alignment can itself be derived from packet-normalized theta-source equality and calibration of the upper-semi target with the packet-normalized value.

135. The entry-target calibration can likewise be expressed by equality to the packet-normalized value, deriving equality with the upper-semi target.

136. The Step (xi-f) real algebra proves $C_\Theta \geq -1$, strictness under strict upper-ray membership, and equality constraints in the boundary case.

137. The final route rejects the balanced sign-compatible comparison level as sufficient; it requires hull/log-volume comparison data.

138. The local Frobenioid shift model is now sourced by a p_v^N log-volume action package and a global realified calibration package; a non-automorphic local shift with nonzero step is separated from the calibrated value, and the upper-ray quotient experiment gives the exact threshold for collapse versus separation in the boundary upper ray.

139. Several downstream IUT IV numerical estimates are formalized as ordered-real algebra conditional on the Corollary 3.12 input.

The first-pass experiment report evaluates the route-theorem, finite Gaussian, collapse-obstruction, upper-ray, and IUT IV algebra flags to `true`. The flag `disputeSettledByCurrentStage` remains `false`.

The current code base is large enough to distinguish three meanings of "available". First, pure ordered-real consequences are proved outright. Second, route theorems prove that certain named source-facing records imply the desired endpoint. Third, the records themselves remain obligations when they stand for Hodge-theater, Frobenioid, log-Kummer, hull, determinant, or multiradial constructions not yet formalized from IUT I–III. Only the first two meanings are currently achieved.

8 What Is Not Yet Proved

No formal statement currently proves IUT III, Corollary 3.12 from Mochizuki's published hypotheses. The following mathematics is still external to the code:

1. initial Θ-data and its arithmetic hypotheses from IUT I;
2. Hodge theaters, Frobenioids, log-shells, prime strips, and theta links;
3. the Hodge–Arakelov theta evaluation and Gaussian monoids of IUT II;
4. the construction of the actual (Ind1), (Ind2), (Ind3) data in Theorem 3.11;
5. the genuine holomorphic hull and determinant constructions used in Step (xi), beyond the present source-data constructor and ordered-real upper-ray model;
6. the full proof that the typed SHE and Hodge-descent transport obligations follow from Mochizuki's source constructions, beyond the current Hodge–Arakelov canonical-one synchronization shadow;
7. the construction of the input prime-strip link (IPL), simultaneous holomorphic expressibility (SHE), and algorithmic parallel transport (APT) properties as actual consequences of Theorem 3.11 rather than records;
8. the arithmetic-divisor, height, log-different, and log-conductor constructions behind the IUT IV Diophantine applications.

9 Implementation Audit

The code contains no `sorry`, `admit`, or Lean axioms in the Stage 1 corridor. The current risk is not hidden unsoundness of this form, but mathematical incompleteness: several records are deliberately named obligation interfaces. These interfaces are acceptable only as temporary boundaries. The next phase must replace them by constructions from the source papers or mark them as unresolved.

The endpoint type `Stage1Comparison` is not a source proof: its comparison field is the Corollary-3.12-shaped conclusion, and the corresponding theorem is only a projection. The source-facing route must construct this endpoint from the ledger. The ledger now records the opaque output certification and ties the Θ-common bound used in the final inequality to `chartedContainer.apply certificate`, so the displayed charted container cannot be paired with an unrelated convenient bound. Throughout the current corridor, "upper ray" means the signed-log region $\{x \mid x \leq b\}$, not the ordinary order-theoretic ray $\{x \mid b \leq x\}$.

The largest engineering issue is `Iut/Stage1/IUTStage1Source.lean`, which currently contains most of the finite-label, Gaussian, Hodge/SHE, hull, Step (x), Step (xi), Step (xii), and IUT IV algebra. This is now too large for comfortable review. Low-risk future refactors are:

finite labels and q^{j^2} averages	separate module,
Step (xi) hull/determinant upper rays	separate module,
Step (xii) local Frobenioid shifts	separate module,
IUT IV real algebra	separate module.

These refactors should be mechanical only: move declarations, preserve names where possible, and rebuild after each move.

10 Next Work

The next mathematical milestone is to construct, rather than assume, more of the source-facing SHE and Hodge-descent preservation data. The immediate route is:

$$\begin{gathered}
q^{j^2} \text{ finite Gaussian endpoint with pivotal restriction} \\
\Downarrow \\
\text{Gaussian and splitting monoid construction} \\
\Downarrow \\
\text{factored square/full-label preservation} \\
\Downarrow \\
\text{Hodge/SHE transport audit} \\
\Downarrow \\
\text{hull/log-volume comparison} \\
\Downarrow \\
C_\Theta \geq -1.
\end{gathered}$$

The project should continue by replacing the present record fields for factored SHE preservation and Hodge-descent source-target alignment with the corresponding constructions from IUT I–III.

The practical priority order is:

1. factor the large source module into reviewable finite-label, hull, Frobenioid-shift, and IUT IV algebra modules without changing theorem statements;

2. extend the Gaussian-derived factored SHE construction beyond the finite degree-evaluation and canonical-label shadows toward actual Gaussian/Frobenioid material from IUT II;

3. construct the packet-correspondence source from the actual log-Kummer/Frobenioid machinery of IUT III, Step (x), rather than treating this finite correspondence object itself as source-facing input;

4. replace the hull/determinant endpoint records by the actual holomorphic-hull and determinant operations from Remark 3.9.5 and Step (xi);

5. only then evaluate whether the Stage 1 interface has enough source content to bear on the Mochizuki–Scholze–Stix disagreement.

References

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[4] S. Mochizuki, *Inter-universal Teichmuller Theory IV: Log-volume Computations and Set-theoretic Foundations.*

[5] S. Mochizuki, *On the Formalization of IUT: Preliminary Progress Report*, 2026.

[6] P. Scholze and J. Stix, *Why abc is still a conjecture.*